

Trout++

Robust Asynchronous Two-Round ECDSA for Arbitrary Thresholds

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What's in a standard? (Theoretically)

- ▶ Formal Description and Security Proofs
- ▶ Technical Specification
- ▶ Reference Implementation

What's in a standard? (Practically)

- ▶ Formal Description and Security Proofs
 - ▶ Presentation of background
 - ▶ Justifications for the cryptographic assumptions
 - ▶ Exceptionally clear to review security proofs
- ▶ Technical Specification
 - ▶ Underlying cryptographic primitives
 - ▶ Algorithms composing the protocol itself
 - ▶ Exact wire specification and test vectors
 - ▶ Commentary on alternative algorithms and practical considerations, implementation advice
- ▶ Reference Implementation
 - ▶ A correct implementation of everything specified
 - ▶ Easy to read and compare against
 - ▶ Follows best practices for safety and could be deployed
 - ▶ Demonstrates the protocol's performance and practicality

What's in a standard? (Practically)

► Methods of Deployment

- Does the protocol guarantee delivery of the output (is it robust)?
- Does the protocol require a broadcast channel? Does it detail how to implement one?
- Does the protocol require agreement on an Asynchronous Common Subset (ACS)? Is such a specification available? What is the complexity of the combined protocol?
- If the protocol aborts, is any information from the setup revealed to the adversary?

Trout++ & ROAST

- ▶ A complete threshold ECDSA signing service which aims to respond to all of the above
- ▶ Requires no setup other than the output of a distributed key generation (DKG) protocol
- ▶ Supports arbitrary thresholds and does not require an honest majority
- ▶ Guarantees output of a signature so long as a threshold is honest and online
- ▶ Works over an asynchronous network, with nothing other than authenticated peer-to-peer channels

- ▶ A two-round threshold ECDSA signing protocol for arbitrary thresholds
- ▶ Supports identifying faulty participants upon abort
- ▶ Supports performing the first round without any context (presigning)
- ▶ Constant per-participant upload in the bulletin board model
- ▶ Constant per-participant prover complexity
- ▶ Linear per-participant verifier complexity (not vulnerable to timing analysis and can be done in variable-time without a loss in security)
- ▶ Constant-time reference implementation in Rust
- ▶ Composable with ROAST

ROAST

- ▶ Straight-forward transform for eligible synchronous two-round threshold signing protocols into robust asynchronous threshold signing protocols
- ▶ Intended for use with FROST, a leading threshold Schnorr signature scheme
- ▶ Only requires authenticated peer-to-peer channels

Trout++ is the first threshold ECDSA signing protocol compatible with ROAST, and their composition is the first robust asynchronous threshold ECDSA signing scheme without an honest majority assumption.

Practicality

- ▶ Complete presentation comprehensive to all considerations
- ▶ Requires the Decisional Diffie-Hellman, Hard Subgroup Membership¹, and Adaptive Root assumptions over a class group of unknown order, with a known-order subgroup where the discrete-log problem is easy²
- ▶ Planning full specification of all class group algorithms
- ▶ $\approx 1.8\text{-}4.6$ KB per-participant upload per Trout++ invocation³
- ▶ $\approx 4.5\text{-}14$ seconds to run the prover in constant-time⁴
- ▶ $\approx 160\text{-}450$ milliseconds to run the prover in variable-time⁵

Benchmarks performed on a single-thread of an M2 Pro CPU.

¹<https://eprint.iacr.org/2018/791>

²<https://eprint.iacr.org/2015/047>

³Without Identifiable Aborts and With a Prior One-Round Setup -
With Identifiable Aborts and Without Any Prior Additional Setup

⁴Portable code compiled with target-specific optimizations

⁵BICYCL (premised on GMP) compiled with target-specific optimizations

Publications

- ▶ Trout++ was accepted to ESORICS, 2026
- ▶ The full paper, with security proofs, will be published to the IACR ePrint server shortly
- ▶ The current implementation, and conference paper, is publicly available now @ <https://github.com/kayabaNerve/trout>

Ongoing Work

- ▶ Accumulating decades of miscellaneous literature into a complete reference for engineers (not academics) to implement
- ▶ Continuing specification of each underlying algorithm, ideally including the complete proofs on their memory usage and termination bounds
- ▶ Further optimizations to the constant-time prover, which does demonstrate a severe gap between the average-case and worst-case termination bounds
- ▶ If time allows, specifying not just Trout++, as composable with ROAST, but also providing a reference implementation and exact specification for the ROAST transformation

Optimized Scaled Decryption

- ▶ Optimized Scaled Decryption is a two-round protocol to simultaneously scale and decrypt a ciphertext, without revealing the scalar or original plaintext.
- ▶ We work with a class group (whose setup is transparent) where G is a uniformly-sampled generators of unknown-order, H is a generator of order q where the discrete-log problem is easy, and $\gcd(\text{ord}(G), q) = 1$.
- ▶ We commit to the scalar, and create the ciphertext to the value, in round one.
 - ▶ Sample $\alpha, \beta \in \mathbb{Z}$, $a, b \in \mathbb{F}_q$
 - ▶ Publish $A = \alpha \cdot G + a \cdot H$
 - ▶ Publish $B = \beta \cdot (q \cdot G) + b \cdot G$
- ▶ We scale and decrypt in round two.
 - ▶ Publish $F = ((\beta * q) + b) \cdot A - \alpha \cdot B = (a * b) \cdot H$

1.
 - ▶ Sample $k, u \in \mathbb{F}_q$
 - ▶ Publish $R' = k \cdot E$
 - ▶ Publish Optimized Scaled Decryption's commitment B for the scalar u
 - ▶ Create ciphertexts N', D' for x, k respectively
2.
 - ▶ Sample randomizers $\rho, \mu \in \mathbb{F}_q$ from the transcript
 - ▶ Set $R = \rho \cdot R' + \mu \cdot E^6$
 - ▶ Derive ciphertexts N, D for $m + R_x * x, \rho * k + \mu$ respectively
 - ▶ Perform Optimized Scaled Decryption's second round with N, B to receive n , then with D, B to receive d
 - ▶ Output $R_x, n * d^{-1}$ as the ECDSA signature

⁶Technique from <https://eprint.iacr.org/2024/1950>